From Absolute Scores to Pairwise Preferences: Training LLM Judges for Chinese Text Rewriting Evaluation

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Abstract

Evaluating Chinese text rewriting is inherently subjective: a good rewrite must preserve meaning while improving expression, and annotators often make relative rather than calibrated absolute judgments. This raises a central question: what supervision signal should be used to train LLM-based rewrite judges? We compare absolute-score supervised fine-tuning (SFT) with pairwise preference training for Chinese rewrite quality evaluation. Using 730 human-rated Chinese source–rewrite pairs from three trained annotators (ICC (2,1) = 0.87), we construct pairwise preference supervision from holistic human ratings and train RewriteJudge to compare and rank rewrites. Pairwise preference training achieves the highest ranking alignment ($\rho = 0.688$), outperforming absolute-score SFT ($\rho = 0.645$) and substantially exceeding zero-shot and general-purpose LLM judges. Data efficiency analysis shows that pairwise supervision maintains stable performance with only 25% of training data. We further show that source-similarity metrics are negatively aligned with human judgments, confirming that rewrite evaluation requires learning human preferences rather than measuring source overlap. Our results suggest that contrastive preference supervision is a more suitable training paradigm for subjective rewriting evaluation than absolute-score calibration.

1 Introduction

A rewrite should be semantically faithful but expressively different. We focus on *general-purpose Chinese text improvement rewriting*, where the goal is to improve clarity, fluency, and readability while preserving the source meaning. Evaluating such subjective transformations is challenging: annotators often make relative quality judgments rather than calibrated absolute scores, and standard metrics such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004) rely on source overlap, which can be negatively aligned with rewrite quality. This raises a central question: *how should we train LLM-based judges for subjective rewriting evaluation, where absolute score calibration is difficult but relative preference is more natural?*

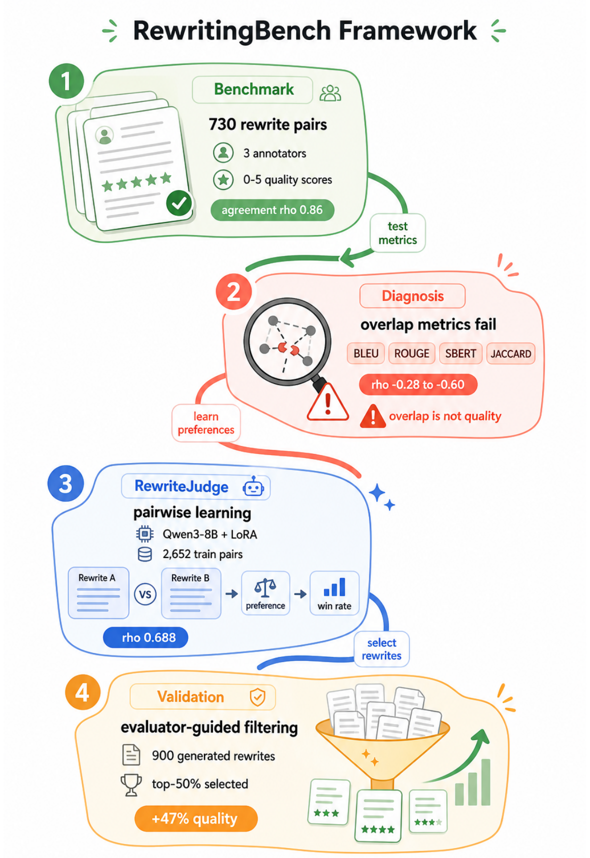

Figure 1: Overview of our approach: we collect human-rated Chinese source–rewrite pairs, construct pairwise preference supervision from holistic human ratings, and train RewriteJudge via contrastive preference learning with win-rate aggregation.

We study two training paradigms for LLM-based rewrite judges: *absolute-score SFT*, which trains the model to predict a 0–5 score directly, and *pairwise preference training*, which trains the model to compare candidate rewrites and aggregates pairwise preferences into a global ranking. Using 730 human-rated Chinese source–rewrite pairs (Figure 1), we show that pairwise preference training better aligns with human rankings than absolute-score SFT and substantially outperforms zero-shot and general-purpose LLM judges. We further provide diagnostic evidence that source-similarity metrics are negatively aligned with human judgments, motivating the need for learned preference-based evaluation.

Our contributions are:

1. We formulate Chinese rewriting evaluation as a preference-learning problem, where the goal is to align LLM judges with human quality rankings rather than merely predict calibrated absolute scores.
2. We propose a pairwise preference training framework for RewriteJudge, converting human holistic ratings into pairwise comparisons and aggregating model preferences into global quality rankings via win rates.
3. We empirically show that pairwise preference training outperforms absolute-score SFT and zero-shot/general-purpose judges on Chinese rewriting evaluation ($\rho = 0.688$ vs. $\rho = 0.645$ for absolute SFT).
4. We provide diagnostic evidence that source similarity is a poor proxy for rewrite quality, motivating the need for learned preference-based evaluation.

2 Related Work

2.1 Source Similarity and Reference-Based Metrics

Traditional rewriting evaluation relies on overlap metrics like BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004), while embedding metrics such as BERTScore (Zhang et al., 2020), MoverScore (Zhao et al., 2019), BARTScore (Yuan et al., 2021), and SBERT (Reimers and Gurevych, 2019) improve semantic matching through contextualized representations. Methods such as ParaScore (Lan and Chen, 2023) and Alam et al. (2024) consider paraphrase quality and multiple dimensions, and prior surveys document broader obstacles in NLG evaluation (Gehrmann et al., 2023; Gao et al., 2025). These methods mostly treat similarity as a useful proxy for quality. We do not claim to exhaust all automatic evaluation methods; instead, we focus on a diagnostic question: whether source-rewrite similarity, a common proxy in rewriting evaluation, aligns with human quality judgments. Our work tests this assumption in Chinese rewriting, where desirable transformation may make similarity and quality conflict.

2.2 LLM-as-a-Judge and Task-Specific Evaluators

The LLM-as-a-Judge paradigm offers a flexible alternative to reference metrics. G-Eval (Liu et al., 2024) uses prompting for structured evaluation, Prometheus 2 (Kim et al., 2024) fine-tunes a 7B model for absolute and pairwise scoring, and Chatbot Arena (Chiang et al., 2024) leverages human pairwise comparisons. However, general-purpose judges can exhibit bias and rating inconsistency (Wang et al., 2024a,b; Lee et al., 2025). We examine whether such judges transfer to the specialized, subjective task of Chinese rewrite quality evaluation and whether task-specific training is necessary.

2.3 Pairwise Preference and Ranking-Based Evaluation

Pairwise comparison is often more stable than absolute scoring because it asks for a relative judgment rather than global calibration. Kim et al. (2024) find higher agreement with human judgments in general NLP tasks, Zheng et al. (2023) use pairwise comparison for LLM assessment, and work on generative selection shows that comparative scoring can improve selection from N candidates (Wang et al., 2024c). Preference learning work further shows that models can learn from pairwise signals without explicit reward scores (Rafailov et al., 2023; Lambert et al., 2024). We test whether this comparative signal specifically helps Chinese rewriting in a cross-source global ranking setting, where human quality judgments are subjective and ordinal.

3 Method

Given a Chinese source text s and a candidate rewrite r, the evaluator should judge whether r improves expression while preserving the source meaning. A high-quality rewrite should be faithful, fluent, natural, and appropriately transformed. We study two paradigms for training LLM-based judges (Figure 2).

Absolute-score SFT. The standard approach trains a model to predict a holistic quality score:

$$f_\theta : \mathcal{S} \times \mathcal{R} \to [0, 5] \tag{1}$$

The training objective minimizes the discrepancy between predicted and human-assigned scores:

$$\mathcal{L}_{\text{SFT}} = \mathbb{E}_{(s,r,q)} \left[\ell(f_\theta(s, r),\, q) \right] \tag{2}$$

where q is the human quality rating and ℓ is a cross-entropy loss over discrete score classes.

Pairwise preference training. Rather than predicting absolute scores, we train the model to compare pairs of rewrites:

$$g_\theta : (\mathcal{S} \times \mathcal{R})^2 \to \{0, 1\} \tag{3}$$

Given two source–rewrite pairs (s_i, r_i) and (s_j, r_j) with human ratings $q_i > q_j$, the model learns:

$$\mathcal{L}_{\text{pair}} = \mathbb{E}_{(r_i, r_j)} \left[-\log P_\theta(r_i \succ r_j \mid q_i > q_j) \right] \tag{4}$$

At inference, pairwise preferences are aggregated into a global ranking via win rates (Eq. 6). The key difference is that absolute scoring requires global calibration across all quality levels, while pairwise training only requires local relative judgments.

3.1 Human Preference Data

3.1.1 Quality Dimensions and Annotation

We define text rewriting quality along four complementary dimensions, scored from 0 to 5: **Meaning preservation** (preserve the source meaning without factual errors or omissions of key content), **Expression improvement** (improve fluency, clarity, and readability), **Appropriate transformation** (make necessary and effective structural, phrasing, and syntactic adjustments, rather than mechanical substitution or excessive rewriting), and **Naturalness and idiomaticity** (ensure the Chinese expression is natural and contextually appropriate). The final score is a holistic judgment integrating all dimensions; annotators provide only a unified rating, using the four dimensions as internal guidelines. We do not reward divergence for its own sake: a rewrite that is lexically or structurally different but semantically distorted or less natural should receive a low score.

3.1.2 Data Collection

We collected 730 Chinese rewrite pairs (730 unique source texts). Source texts contain 49 to 251 Chinese characters (mean 103.2, median 93), while rewrites contain 20 to 247 characters (mean 93.4, median 84.5). Three trained annotators independently scored each pair on the holistic scale from 0 to 5. Agreement is high across multiple statistics: mean pairwise Spearman

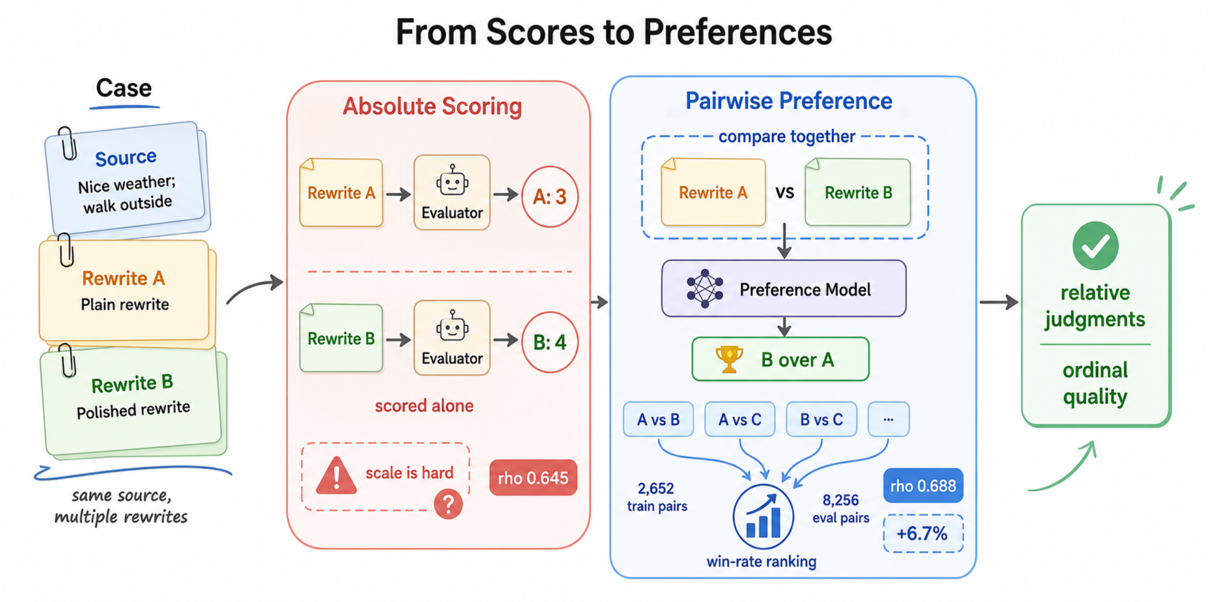

Figure 2: Conceptual illustration of absolute scoring and pairwise preference evaluation. Absolute scoring judges each rewrite independently on a fixed scale, whereas pairwise preference evaluation compares candidate rewrites jointly and aggregates comparison outcomes into a win rate ranking. The pairwise setting in our benchmark compares source-rewrite pairs across different sources for global quality ranking.

$\rho = 0.88$, ICC(2,1) = 0.87, Krippendorff's alpha (ordinal) = 0.87, mean weighted kappa (quadratic) = 0.87, exact agreement = 81.1%, and ± 1 agreement = 94.1%. We use the average rating as the gold standard. The dataset is split into 600 training pairs and 130 evaluation pairs (129 after removing one parsing issue) using stratified sampling by score. The evaluation set closely reflects the full score distribution: 0 (15.5%), 1 (27.9%), 2 (24.8%), 3 (20.2%), 4 (9.3%), 5 (2.3%).

3.1.3 Pair Construction for Preference Training

From the 600 training samples, we construct pairwise comparisons for preference training. For each pair, the two rewrites must differ by at least 0.67 points in human score:

$$\mathcal{P} = \{(r_i, r_j) \mid |q_i - q_j| \geq 0.67\} \qquad (5)$$

yielding 2,652 balanced comparisons (1,326 per class). Because each source has one rewrite, pairs are constructed across different source texts. The model receives both source–rewrite pairs and chooses which demonstrates higher overall quality relative to its own source.

3.1.4 Class Balancing

The score distribution is imbalanced (Score 0: 15%, Score 5: 3%). When we train RewriteJudge on the original data, 82% of its predictions collapse to Score 1 and the correlation with human scores is low ($\rho = 0.253$). We therefore oversample minority score classes and create a balanced training set of 1,008 samples (168 per class). After balancing, the model predicts across the full score range and better matches human rankings.

3.2 RewriteJudge Training

3.2.1 Model Architecture and Training

We fine-tune Qwen3 8B (Yang et al., 2025) and Qwen2.5 7B Instruct (Team, 2025) with LoRA (Hu et al., 2022). For absolute scoring, the model outputs a single holistic score from 0 to 5 using a rubric-grounded prompt. For pairwise preference learning, the model receives both source–rewrite pairs and chooses which candidate demonstrates higher quality. Full training details and hyperparameters are provided in Appendix A.

3.2.2 Rubric-Grounded Prompting

We compare three prompt variants for absolute scoring: **Simple** (minimal instructions), **Standard** (brief criteria), and **Detailed**

Table 1: Main comparison: pairwise preference training outperforms absolute-score SFT and zero-shot judges on Chinese rewriting evaluation.

Training paradigm	Model	Supervision	Spearman ρ
Pairwise (ours)	**Qwen3 8B**	**Cross-source pref.**	**+0.688**
Pairwise (ours)	Qwen2.5 7B	Cross-source pref.	+0.665
Absolute SFT (ours)	Qwen3 8B	Score regression	+0.645
Absolute SFT (ours)	Qwen2.5 7B	Score regression	+0.580
Zero-shot	Qwen2.5 72B	None	+0.406
Zero-shot	DeepSeek V3	None	+0.391
Gen. purpose	Prometheus 2	Multi-task	+0.145
Zero-shot	Qwen3 8B	None	+0.106

(matching the human annotation rubric). The Detailed prompt grounds the model directly in the human annotation protocol (see Appendix H). Rubric-aligned prompts improve absolute scoring ($\rho = 0.645$ vs. $\rho = 0.607$ for Standard), and reasoning prefixes hurt performance ($\rho = 0.360$).

3.2.3 Win-Rate Aggregation

At inference, all $\binom{129}{2} = 8{,}256$ evaluation pairs are scored; each rewrite's *win rate* is:

$$\text{WinRate}(r_i) = \frac{1}{N-1} \sum_{j \neq i} \mathbf{1}[\text{Pref}(r_i, r_j) = 1] \tag{6}$$

Spearman correlation is then computed between win rates and human average scores. This aggregation provides a more robust quality estimate than any single absolute prediction, as each rewrite's ranking is informed by multiple comparisons.

4 Experiments

4.1 Pairwise Preference Training Outperforms Absolute SFT

Table 1 presents the central comparison. Cross-source pairwise preference training achieves $\rho = 0.688$ on Qwen3 8B, outperforming absolute-score SFT ($\rho = 0.645$) by 0.043 in Spearman ρ. The same pattern holds on Qwen2.5 7B ($\rho = 0.665$ vs. $\rho = 0.580$, a gain of 0.085). Both trained variants substantially exceed zero-shot and general-purpose judges, where even 685B-parameter models fail to reach positive correlation in some cases.

4.2 Comparison with All Baselines

Table 2 provides the full comparison across 19 methods. Source-similarity metrics are all negatively correlated with human scores ($\rho = -0.28$ to $\rho = -0.60$). Among zero-shot LLMs, Qwen2.5 72B ($\rho = 0.406$) and DeepSeek V3 ($\rho = 0.391$) show moderate positive correlation, while several larger or newer models perform poorly: GPT-4.1 ($\rho = -0.145$), DeepSeek V3.2 ($\rho = -0.274$), Kimi K2 ($\rho = -0.472$). Prometheus 2, despite being trained for evaluation, reaches only $\rho = 0.145$, confirming that general-purpose evaluator training does not transfer to Chinese rewriting.

4.3 Cross-Source Analysis and Why Pairwise Helps

Under the Qwen2.5 7B setting, cross-source pairwise comparison reaches $\rho = 0.665$, while same-source preference yields $\rho = 0.421$. Cross-source comparisons pair rewrites from different source texts and therefore cover a wider quality range, but they require a global notion of quality and are less directly aligned with the practical setting of choosing among rewrites for the same input. Same-source preference evaluation remains an important open direction.

We attribute the pairwise advantage over absolute scoring to two factors: (1) pairwise comparison provides an easier local learning signal than direct score calibration, although cross-source comparison still requires a global notion of quality; (2) win rate aggregation provides a more robust quality estimate than any single absolute prediction, as each rewrite's score is informed by multiple comparisons. This improvement comes at the cost of $O(n^2)$ comparisons at inference.

4.4 Analysis

4.4.1 Data Efficiency

Table 3 shows pairwise performance with different training data fractions. Qwen3 8B reaches $\rho = 0.611$ with only 25% of the data (663 pairs) and $\rho = 0.669$ with 50%, demonstrating that pairwise supervision is data-efficient and maintains stable performance even with limited training data.

Qwen2.5 7B shows an irregular result at 50% data caused by position bias, where the model frequently prefers the second presented rewrite regardless of content. Position swap averaging mitigates this bias by presenting each pair in both orderings.

Table 2: Full evaluation results on the Chinese rewriting quality benchmark. Source-similarity metrics are negatively correlated with human scores. Best results are in **bold**.

Method	Type	Spearman ρ	Pearson r	Kendall τ	MAE	RMSE	Exact / ±1 / ±2 (%)
RewriteJudge (Ours, Qwen3 8B)	**Fine tuned (8B)**	**+0.645**	**+0.602**	**+0.530**	**0.91**	**1.24**	25.6 / 79.8 / 95.4
+Reasoning prefix	Fine tuned (8B)	+0.360	+0.374	+0.312	1.07	1.36	14.0 / 72.1 / 93.0
Unbalanced training (Qwen3 8B)	Fine tuned (8B)	+0.337	+0.344	+0.289	1.12	1.43	14.0 / 70.5 / 90.7
RewriteJudge (Ours, Qwen2.5 7B)	Fine tuned (7B)	+0.580	+0.580	+0.470	0.94	1.25	24.8 / 75.2 / 93.0
Unbalanced training (Qwen2.5 7B)	Fine tuned (7B)	+0.287	+0.246	+0.238	1.10	1.40	14.0 / 72.1 / 91.5
Prometheus 2	Fine tuned (7B)	+0.145	+0.126	+0.120	1.54	1.95	17.0 / 53.8 / 75.5
Zero shot (Qwen3 14B)	Zero shot (14B)	+0.129	+0.180	+0.112	2.03	2.37	10.9 / 35.7 / 56.6
Zero shot (Qwen2.5 14B)	Zero shot (14B)	+0.058	+0.105	+0.046	2.14	2.49	11.6 / 30.2 / 56.6
Zero shot (Qwen3 8B)	Zero shot (8B)	+0.106	+0.176	+0.093	1.86	2.23	15.5 / 40.3 / 62.8
Zero shot (Qwen2.5 7B)	Zero shot (7B)	−0.019	+0.079	−0.016	2.04	2.38	10.1 / 37.2 / 56.6
G-Eval (Qwen3 8B)	Zero shot (8B)	+0.041	+0.089	+0.033	1.85	2.22	16.3 / 41.1 / 58.1
G-Eval (Qwen2.5 7B)	Zero shot (7B)	−0.145	−0.064	−0.121	2.22	2.62	12.4 / 33.3 / 52.7
SBERT COSINE	Traditional	−0.280	−0.072	−0.199	NA	NA	NA
BLEU	Traditional	−0.294	−0.211	−0.243	NA	NA	NA
W2V COSINE	Traditional	−0.336	−0.099	−0.238	NA	NA	NA
ROUGE L	Traditional	−0.385	−0.396	−0.317	NA	NA	NA
JACCARD WORD	Traditional	−0.538	−0.509	−0.416	NA	NA	NA
TFIDF COSINE	Traditional	−0.571	−0.503	−0.433	NA	NA	NA
JACCARD CHAR	Traditional	−0.595	−0.550	−0.453	NA	NA	NA

Table 3: Data efficiency: pairwise performance at different training data fractions (LoRA $r = 16$, 3 epochs).

Model	Training Data	Pairs	Spearman ρ
Qwen2.5 7B	25%	663	+0.434
	50%	1,326	+0.140
	100%	**2,652**	**+0.665**
Qwen3 8B	25%	663	+0.611
	50%	1,326	+0.669
	100%	**2,652**	**+0.688**

Table 4: Pairwise accuracy and ranking correlation are poorly aligned for zero-shot models.

Model	Accuracy (%)	Spearman ρ	Paradox?
Ours (8B LoRA)	69.5	+0.688	No
Ours (7B LoRA)	90.0	+0.665	No
Qwen3 235B	72.0	+0.132	Yes
Qwen2.5 72B	69.3	+0.406	Mild
DeepSeek V3	67.7	+0.391	No
Kimi K2	31.6	−0.472	No

4.4.2 Accuracy vs. Ranking Correlation

A natural expectation is that higher pairwise accuracy should produce higher ranking correlation. Our results contradict this (Table 4): Qwen3 235B achieves 72.0% pairwise accuracy yet only $\rho = 0.132$, while our fine-tuned 8B model reaches $\rho = 0.688$ at 69.5% accuracy. This suggests that zero-shot models may correctly identify easy pairs with large quality gaps but fail to maintain consistent logic across the full spectrum.

Table 5: Bias analysis. Lower absolute correlation values indicate less bias. RewriteJudge shows low bias across all dimensions compared to baselines.

Method	Output Len. r	Input Len. r	Verbosity ρ	Position Bias r	Bias Score
RewriteJudge (Qwen3 8B)	-0.033	-0.070	0.006	0.120	0.057
RewriteJudge (Qwen2.5 7B)	-0.058	-0.115	0.106	0.187	0.116
Prometheus 2	0.010	-0.071	0.098	0.223	0.100
Zero-shot Qwen3 8B	0.325	0.167	0.382	0.437	0.328
Char Overlap	0.306	0.044	0.760	—	0.468
Length Heuristic	0.287	0.052	0.741	—	0.441

4.4.3 Training Robustness

Training on the imbalanced score distribution causes mode collapse (Qwen3 8B: $\rho = 0.337$), while class-balanced oversampling (168 per score class) improves performance to $\rho = 0.645$, confirming that rewrite evaluation requires supervision across the full quality range. Additional ablations (LoRA rank sensitivity) are reported in Appendix B.

4.4.4 Bias Checks: Length, Verbosity, and Position Effects

A reliable evaluator should judge quality based on content, not spurious features such as text length or verbosity. Table 5 compares bias across methods. Both RewriteJudge variants show minimal output length bias (Qwen3 8B: $r = -0.033$; Qwen2.5 7B: $r = -0.058$), well below zero-shot Qwen3 8B ($r = 0.325$), character overlap ($r = 0.306$), and a length heuristic ($r = 0.287$). RewriteJudge Qwen3 8B achieves near-zero verbosity bias ($\rho = 0.006$), far below zero-shot Qwen3 8B ($\rho = 0.382$) and character overlap ($\rho = 0.760$).

Position bias measures candidate-order preference in pairwise evaluation. RewriteJudge

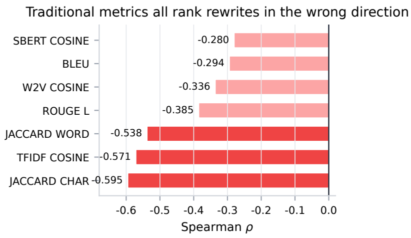

Figure 3: Source-similarity metrics are negatively aligned with human rewrite quality. All seven metrics show negative Spearman ρ with bootstrap 95% CIs that do not cross zero.

maintains low position bias (Qwen3 8B: $r = 0.120$; Qwen2.5 7B: $r = 0.187$), while zero-shot Qwen3 8B shows higher bias ($r = 0.437$). Extended bias analysis with additional figures is provided in Appendix D.

4.5 Diagnostic Analysis: Source Similarity Fails

Before training task-specific judges, we ask whether automatic metrics based on source similarity can serve as proxies for rewrite quality. Figure 3 shows that all seven source-similarity metrics are negatively correlated with human judgment ($\rho = -0.28$ for SBERT COSINE to $\rho = -0.60$ for JACCARD CHAR), with bootstrap 95% confidence intervals that do not cross zero. This occurs because quality rewrites often diverge from the source wording, which overlap metrics penalize.

Two systematic error patterns explain this failure: (1) *surface similarity without quality*—rewrites that are close copies receive high overlap scores but low human ratings because little rewriting occurred; (2) *quality without surface similarity*—strong rewrites that restructure syntax and vary vocabulary receive low overlap scores, even when they preserve meaning and improve fluency. These findings motivate the need for learned preference-based evaluation rather than source-similarity proxies. Additional error analysis with concrete examples is provided in Appendix G.

4.6 Discussion and Limitations

RewriteJudge substantially improves ranking alignment over zero-shot and general-purpose evaluators, and bias analysis confirms it is not driven by length or verbosity features. However, this result demonstrates that task-specific supervision helps for this benchmark and task definition, not that RewriteJudge generalizes as a universal rewriting evaluator. Cross-source pairwise aggregation achieves the highest correlation ($\rho = 0.688$), but applies to global ranking rather than same-source selection and comes at $O(n^2)$ cost.

Scope and generalization. Our dataset contains 730 Chinese general-purpose improvement pairs from a single-rewrite-per-source design; generalization to other languages, rewriting subtasks (style transfer, simplification), and same-source candidate selection remains unknown.

Validation and calibration. Our analysis focuses on ranking quality; better-calibrated numeric scores are an open direction. Full-parameter tuning and larger annotated datasets may further improve performance beyond the LoRA adaptation used here.

5 Conclusion

We studied how to train LLM-based judges for Chinese text rewriting evaluation, comparing absolute-score SFT with pairwise preference training. Using 730 human-rated Chinese source–rewrite pairs, we showed that pairwise preference training achieves the highest ranking alignment ($\rho = 0.688$), outperforming absolute-score SFT ($\rho = 0.645$) and substantially exceeding zero-shot and general-purpose LLM judges. Data efficiency analysis demonstrates that pairwise supervision maintains stable performance with only 25% of training data. We further showed that source-similarity metrics are negatively aligned with human judgments, confirming that rewrite evaluation requires learning human preferences rather than measuring source overlap. Our results suggest that contrastive preference supervision is a more suitable training paradigm for subjective rewriting evaluation than absolute-score calibration.

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A Training Configuration Details

Table 6 summarizes the training configuration for RewriteJudge. We employ Low Rank Adaptation (LoRA) to fine-tune the model, injecting trainable matrices into seven target modules: query, key, value, output projection, and the three MLP layers (gate, up, and down projections). This configuration results in trainable parameters totaling approximately 0.2% of the full model.

For a pretrained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, LoRA reparameterizes the update as:

$$W' = W_0 + \frac{\alpha}{r} BA \tag{7}$$

where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$. Only A and B are trained while W_0 remains frozen.

A.1 Hyperparameters

The LoRA adapters are configured with rank $r = 16$, scaling factor $\alpha = 32$, and a dropout rate of 0.05. We fine-tune the model for 3 epochs using a learning rate of 2×10^{-4} with a cosine decay schedule (3% warmup). The training uses bf16 mixed precision. With a batch size of 4 and 4 gradient accumulation steps (effective batch size 16), the total training time is approximately 16 minutes on a single NVIDIA H20 GPU.

Table 6: Complete training configuration for RewriteJudge.

Hyperparameter	Value
Base model	Qwen3 8B, Qwen2.5 7B Instruct
Quantization	bf16 (no quantization)
LoRA rank (r)	16
LoRA scaling (α)	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Learning rate	2×10^{-4}
LR scheduler	Cosine (warmup 3%)
Epochs	3
Batch size	4 (gradient accumulation: 4)
Effective batch size	16
Precision	bf16
Gradient checkpointing	Enabled
Training samples	1,008 (168 per class)
GPU	1× NVIDIA H20 GPU (96GB)
Training time	∼16 minutes

B Pairwise Preference Ablations

B.1 LoRA Rank Ablation

Table 7 investigates the sensitivity of pairwise evaluation quality to LoRA rank (r). We use $r = 16$ as the pre-specified setting for all main experiments. The $r = 32$ results are reported only as a sensitivity analysis on the evaluation set and are not used for model selection. Performance is stable across ranks, with $r = 32$ achieving the highest Spearman ρ of 0.705 for Qwen2.5 7B and 0.713 for Qwen3 8B. The narrow range across ranks (maximal $\Delta\rho \approx 0.03$) indicates that pairwise preference learning is robust to adapter capacity.

Table 7: LoRA rank ablation: effect on trainable parameters and Spearman correlation.

Model	Rank r	α	Params	Spearman ρ
Qwen2.5 7B	8	16	~0.1%	+0.685
	16	32	~0.2%	+0.665
	32	**64**	~0.4%	**+0.705**
Qwen3 8B	8	16	~0.1%	+0.679
	16	32	~0.2%	+0.688
	32	**64**	~0.4%	**+0.713**

B.2 Pairwise Data Efficiency

Table 8 provides additional data efficiency details beyond the summary in Section 4. Qwen2.5 7B shows an irregular result at 50% data caused by position bias, where the model frequently prefers the second presented rewrite regardless of content.

Table 8: Pairwise data efficiency: full details at different fractions of training data (LoRA $r = 16$, 3 epochs).

Model	Training Data	Pairs	Spearman ρ
Qwen2.5 7B	25%	663	+0.434
	50%	1,326	+0.140
	100%	**2,652**	**+0.665**
Qwen3 8B	25%	663	+0.611
	50%	1,326	+0.669
	100%	**2,652**	**+0.688**

Position swap averaging mitigates this bias by presenting each pair in both orderings. A model that always predicts the second rewrite as better produces average preference ≈ 0.5 per pair, collapsing the win rate distribution and yielding near zero Spearman correlation.

C Extended Results

C.1 Learning Curve with All Methods

Table 9 and Figure 4 show how evaluation quality changes as training data increases. Performance improves from 50 to 1,008 samples, with a clear jump after 400 samples. With 100 balanced samples, RewriteJudge reaches $\rho = 0.117$, already above all zero-shot baselines and general-purpose evaluators. The upward trend suggests that the model needs enough examples in each score class, approximately 168 samples here, to learn the distinctions required for strong rewriting evaluation.

Table 9: Learning curve comparison between Qwen2.5 7B and Qwen3 8B evaluators.

Samples	ρ (Qwen2.5 7B)	ρ (Qwen3 8B)
50	−0.026	+0.058
100	+0.094	+0.117
200	+0.259	+0.136
400	+0.235	+0.247
1008 (full)	+0.580	+0.645

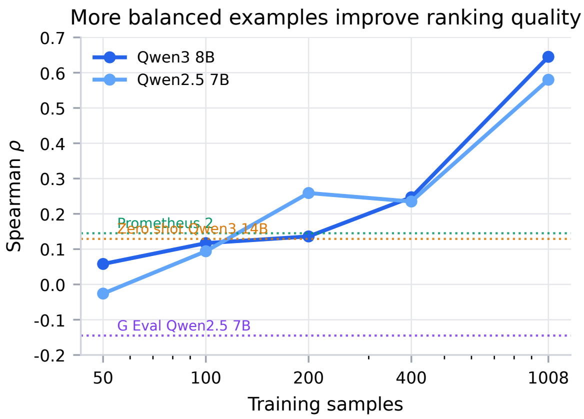

Figure 4: Full learning curve including both LoRA variants and baseline methods (dashed lines). All baselines show flat or negative correlation regardless of data size.

C.2 Score Distribution Analysis

Figure 5 shows the score frequency across different evaluators. Zero shot 7B has a ceiling effect: more than 80% of its predictions are Score 4. Prometheus 2 mostly predicts Score 3 and ignores much of the lower and upper score range. In contrast, RewriteJudge, for both Qwen3 8B and Qwen2.5 7B, produces a distribution closer to the human scores across all six score levels.

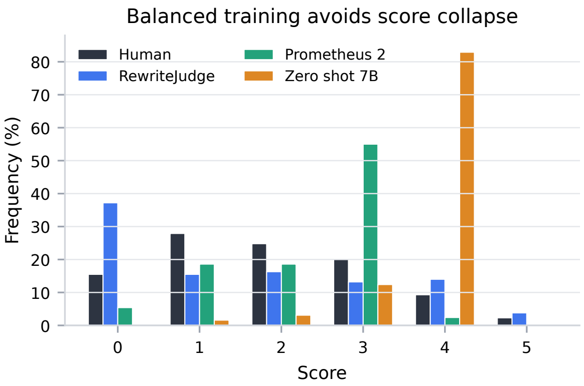

Figure 5: Distribution of human annotations and RewriteJudge predictions. Balanced training makes the model score distribution closer to the human score distribution.

C.3 Confusion Matrix Analysis

Figures 6, 7, and 8 provide a granular view of evaluator errors. Both REWRITEJUDGE variants exhibit a strong diagonal trend consistent with human judgment. The Qwen3 8B variant correctly identifies 60% of Score 0 samples and 66.7% of Score 5 samples, and shows a clear diagonal pattern from upper left to lower right with very few extreme misclassifications, such as predicting Score 5 for a human Score 0. The Qwen2.5 7B variant similarly concentrates predictions along the diagonal: 50% accuracy for Score 0 and 58.3% for Score 4, with the model showing particular strength in the middle and high score range. Both variants rarely commit extreme errors, confirming that class balanced fine tuning preserves ordinal structure.

In contrast, Prometheus 2 has very limited discriminative power: it predicts Score 3 for 63.6% to 73.9% of samples in every human score category. This mode collapse explains its low Spearman correlation (ρ=0.145), as it behaves almost like a constant score predictor.

D Extended Bias Analysis

This section provides additional bias analysis figures and details beyond the summary in Section 4.

D.1 Preliminary Model-Scored Filtering Analysis

As a preliminary use case, we examine whether RewriteJudge can select candidates that receive higher RewriteJudge scores from a generated pool. This experiment is not an independent validation of human-perceived quality. We score 900 generated rewrites (300 sources $\times$ 3 quality levels) with RewriteJudge and then compare several selection rules in Table 10. The numbers in this table are assigned by RewriteJudge itself; they should be read as model scores, not as independent human ratings.

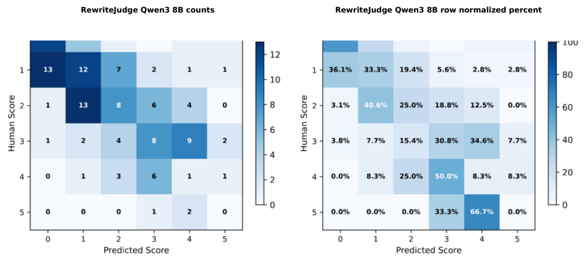

Figure 6: Confusion matrix for RewriteJudge(Qwen3 8B) predictions vs. human average scores (rounded). The model shows strongest agreement for extreme scores, including 0 and 4 to 5.

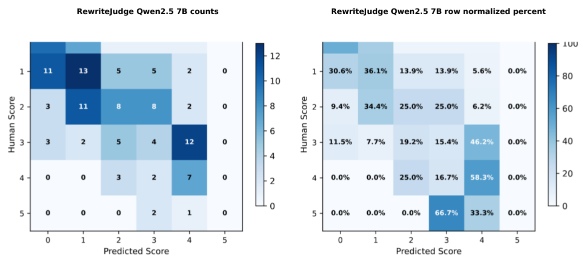

Figure 7: Confusion matrix for RewriteJudge (Qwen2.5 7B) predictions vs. human average scores (rounded). The model shows strongest agreement for extreme scores, including 0 and 4 to 5.

Table 10: Preliminary filtering results scored by RewriteJudge itself. These scores are model outputs, not independent human ratings, and should not be interpreted as human-validated quality gains.

Strategy	N	Mean	σ	%$\geq$3
Top 30%	270	**4.64**	0.48	100.0
$\geq$4 threshold	390	4.44	0.50	100.0
Top 50%	450	4.25	0.67	100.0
$\geq$3 threshold	461	4.22	0.69	100.0
BLEU middle range	868	2.94	1.51	53.1
Random 50%	450	2.90	1.53	51.6
All (unfiltered)	900	2.87	1.54	51.2
Bottom 50%	450	1.48	0.67	2.4

Selecting the top 50% by RewriteJudge score gives a mean model score of 4.25, while a random 50% subset has a mean model score of 2.90. The selected subset has higher model-assigned scores by construction, so this differ-

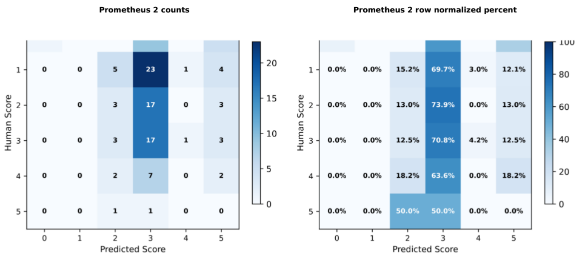

Figure 8: Confusion matrix for Prometheus 2. The model predicts Score 3 for the majority of samples across all human score categories, exhibiting mode collapse that explains its low Spearman correlation (ρ=0.145).

ence should not be interpreted as an independent quality improvement. BLEU-based filtering keeps 96.4% of the dataset and has a mean model score of 2.94, which is close to random selection, consistent with the earlier finding that overlap metrics are weak proxies for rewrite quality. We include two additional reference methods: (1) **character overlap** (JACCARD CHAR), representing traditional overlap metrics; and (2) a **length heuristic** that assigns quality scores based solely on the ratio of rewrite length to source length.

Length bias Figure 9 shows evaluator scores across output length quartiles: both RewriteJudge variants remain nearly flat, while zero-shot Qwen3 8B, character overlap, and the length heuristic all reward longer outputs regardless of quality. The length heuristic exhibits extreme verbosity bias ($\rho = 0.741$), confirming a known failure mode where naive metrics struggle to distinguish between informative expansion and redundant verbosity (Kuang et al., 2024).

Position bias Figure 10 summarizes all four bias dimensions as a compact heatmap: both RewriteJudge variants have low overall bias scores (0.057 and 0.116), while zero-shot Qwen3 8B (0.328), character overlap (0.468), and the length heuristic (0.441) show larger bias values across multiple dimensions.

Verbosity bias. Figure 11 breaks down verbosity bias by the ratio of output length to input length across two categories present in our evaluation set: Concise (ratio <0.8) and Similar (0.8 to 1.2). RewriteJudge variants assign consistently low scores in both categories, remaining well below all other methods in absolute score level. In contrast, other methods show larger gaps: character overlap shows the largest jump (3.4 $\rightarrow$ 4.6, Δ=1.2), reflecting strong systematic preference for longer outputs. The verbosity bias scores in Table 5 (Section 4) confirm this pattern.

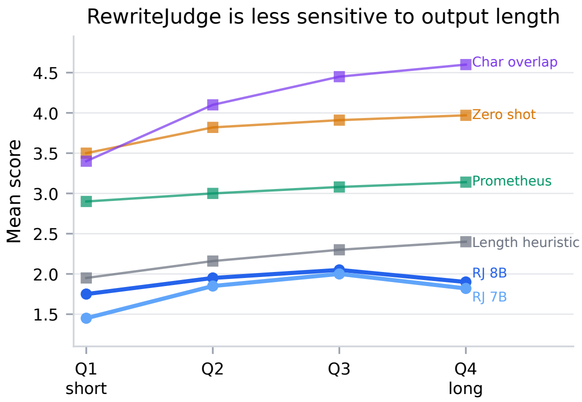

Figure 9: Mean evaluator score by output length quartile. RewriteJudge maintains consistent scores across quartiles, while traditional metrics show increasing scores for longer outputs.

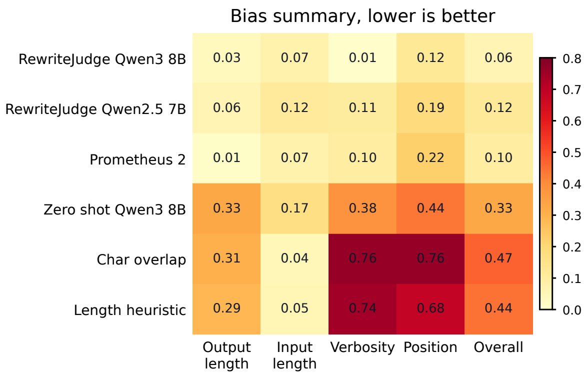

Figure 10: Bias heatmap across evaluator methods. Lower values indicate weaker correlation with spurious factors such as length, verbosity, or position.

E Extended Zero Shot Baseline Analysis

E.1 Accuracy vs. Ranking Correlation

This section provides additional details beyond the summary in Section 4. As visualized in Figure 12, Qwen3 235B achieves high zero-shot accuracy (72.0%) yet fails to provide a meaningful ranking ($\rho = 0.132$). This suggests that zero-shot models may correctly identify easy pairs with large quality gaps but fail to maintain consistent logic across the full spectrum. Our fine-tuned model variants largely avoid this mismatch, as seen by their position in the upper right area of the figure.

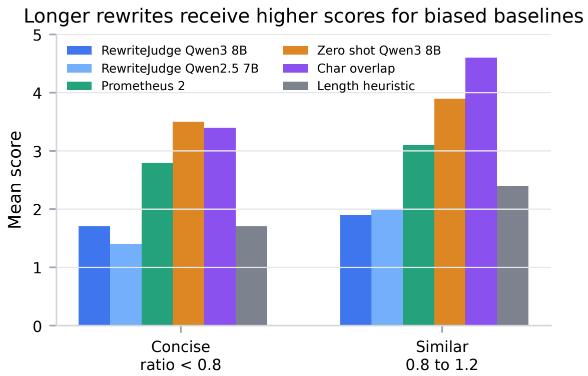

Figure 11: Verbosity bias comparison. RewriteJudge shows almost no verbosity bias, while the length heuristic exhibits extreme preference for verbose outputs (ρ=0.741).

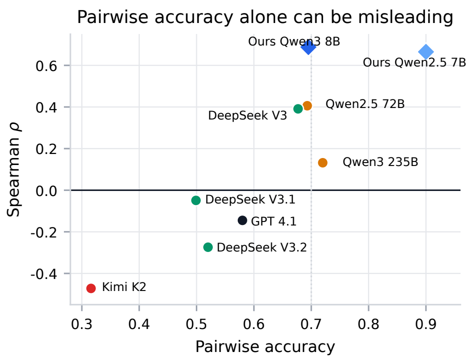

Figure 12: Pairwise accuracy vs. Spearman ρ: high accuracy but low ρ shows that accuracy can mislead for zero-shot evaluators.

E.2 Format Compliance vs. Evaluation Quality

We examine whether format compliance (the ability to follow the pairwise comparison output format) correlates with evaluation quality. Table 11 shows the parse failure rate for each zero-shot baseline.

Table 11: Format compliance vs. evaluation quality. Models with zero parse failures include both the best and worst performers, indicating format compliance is independent of evaluation capability.

Model	Parse Fail (%)	Spearman ρ	Quality
DeepSeek V3	0.0	+0.391	Best zero-shot
Kimi K2	0.0	−0.472	Worst zero-shot
DeepSeek V3.1	0.0	−0.049	Poor
DeepSeek V3.2	0.0	-0.274	Poor
GPT 4.1	0.0	-0.145	Poor
Qwen2.5 72B	8.8	+0.406	Good
Qwen3 235B	9.0	+0.132	Poor

The results in Figure 14 confirm that format compliance and evaluation quality are **orthogonal**. Models with 100% compliance (0% failure) span the entire performance spectrum, from the best zero-shot baseline (DeepSeek V3) to the worst (Kimi K2). This indicates that the mechanical ability to follow instructions is independent of the critical capacity to judge text quality.

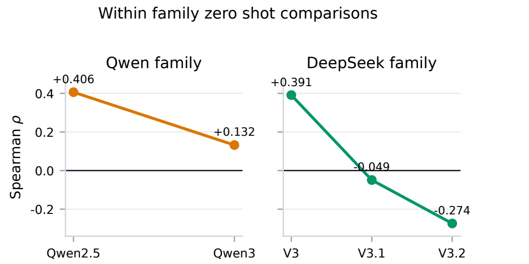

Figure 13: Within-family zero-shot comparisons. These results show that newer or larger endpoints are not reliably better in our prompting setup, but should not be read as a general generational trend.

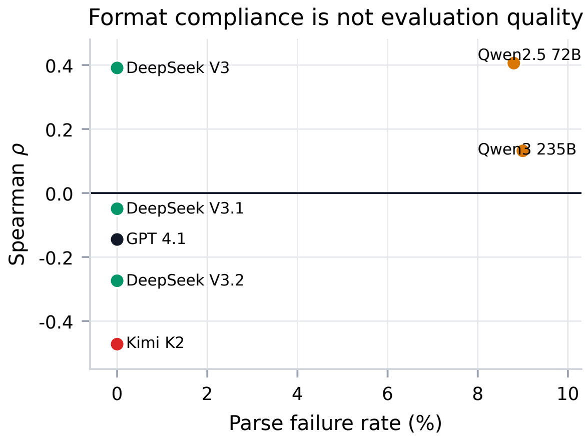

Figure 14: Parse failure rate vs. Spearman ρ. No correlation exists between format compliance and evaluation quality: models with 0% parse failures span the full quality range.

F Chinese Annotation Examples

To provide a qualitative understanding of the benchmark, Table 12 presents several representative cases covering various quality levels. These examples illustrate the nuances of Chinese text rewriting, ranging from simple lexical substitutions to sophisticated stylistic enhancements. We observe that REWRITEJUDGE (RJ) predictions align closely with human gold standard scores, successfully capturing subtle improvements in fluency and expressiveness while remaining sensitive to instances where the rewrite fails to provide significant added value.

Table 12: Representative examples from the evaluation benchmark. **Source**: original text; **Rewrite**: model generated rewrite; **Human**: average human score; **RJ**: RewriteJudge prediction.

Source	Rewrite	Human	RJ
今天天气很好，我们决定去公园散步。	阳光明媚，微风轻拂，我们来到公园漫步，享受这惬意的时光。	4	4
这个问题的解决方案需要进一步研究。	要解决此事还需深入探究才能找到答案。	2	2
他是一个非常好的学生，学习成绩优异。	他学习刻苦努力，是个品学兼优的好学生。	3	3
这家餐厅的菜很好吃。	这儿的菜品味道鲜美可口，令人回味无穷。	3	2
会议上讨论了关于环保的问题。	大家在会议中就环境保护议题展开了深入探讨。	4	4

G Additional Error Analysis: Why Traditional Metrics Fail

Section 4.5 summarizes the core error mechanism in the main text. Here we provide additional detail on the two recurring patterns behind the negative correlation between overlap metrics and human judgment.

G.1 Two Error Patterns

We identify two systematic error patterns that explain the negative correlations:

Pattern 1: Surface similarity without quality. Rewrites that are close copies of the source text, changing only a few words while preserving the original structure, receive high overlap metric scores but low human ratings. For example, a rewrite that substitutes "加盟" with "加盟后" and "场场" with "几乎每场" achieves BLEU=0.55 but receives a human score of only 0.7. The metric rewards lexical overlap that reflects copying, not genuine rewrite quality.

Pattern 2: Quality without surface similarity. Strong rewrites that restructure the sentence, using different vocabulary and syntax while preserving meaning, receive very low metric scores but high human ratings. A rewrite that restructures a passage about a press conference into a more concise summary achieves BLEU=0.00 but receives a human score of 4.0. The metric penalizes the very lexical variation that characterizes successful rewriting.

G.2 Root Cause: Misaligned Objectives

These two patterns show that overlap metrics and human ratings answer different questions: overlap metrics measure how *similar* a rewrite is to its source, while human judgment evaluates how *good* the rewrite is. The critical insight is that these are not merely different metrics; they are *inversely correlated* for rewriting. Pattern 1 shows that high source similarity indicates *insufficient transformation* (the rewrite barely changed anything), while Pattern 2 shows that high source dissimilarity indicates *successful transformation* (the rewrite meaningfully transformed the text). This inverse relationship explains why the negative correlations are universal and statistically significant rather than reflecting random noise or a specific metric deficiency. For text rewriting, a task where the goal is to differ from the source while preserving meaning, these two objectives are inversely correlated. This explains why the strongest negative correlations are observed for character level metrics (JACCARD CHAR: ρ=−0.60), which most directly measure surface overlap, while semantic metrics like SBERT COSINE (ρ=−0.28) show somewhat weaker, but still negative, correlations.

G.3 Implications

This finding has practical implications for Chinese rewriting evaluation: source-rewrite similarity metrics are unreliable proxies for rewrite quality in our benchmark, and task-specific evaluators trained on human-rated data provide a more aligned alternative.

H Scoring Prompts for LLM and Human Evaluation

We investigate how prompt detail affects evaluation quality by comparing three variants: **Simple** (minimal), **Standard** (brief criteria),